



SCHOOL *of* DATA SCIENCE

Troubleshooting

DS 6015: Data Science Capstone
School of Data Science
University of Virginia

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Troubleshooting

Review of common issues encountered by earlier teams

Potential solutions are proposed

Not Enough Time

Q: What if there isn't enough time to complete capstone?

- > Capstone projects are generally scoped to fit
- > Projects kick off the term before to allow for onboarding and discovery.
Teams that use this time wisely tend to do better.
- > If you are waiting on feedback, data, etc., stay productive.
What else can be done?
- > Share the work across students

Unclear Problem Definition

Each project should have a problem statement, but clarity may vary.

Q: What if the sponsor doesn't clearly define the problem?

You might propose ideas at intersection of what's achievable & interesting

Lack of Data

Each project should have data, but coverage and quality may vary

Q: What if the data won't support the objective?

You might propose ideas on what is achievable

- Solve for subset of population?
- Pivot the question?
- If labeled dataset is small, you might train on labeled data, predict on unlabeled, and review cases with high/low probability.
- Can more data be collected? Can supplemental data be created?

Making Meetings more Productive

Each sponsor should meet w students as needed.

Q: What if the meetings aren't productive enough?

You might try:

- Forming an agenda / agreeing on it beforehand
- Sharing results beforehand to discuss during meeting
- Decide Next Steps in the meeting
- Call out decision points in meeting
- Ask for feedback

Ideally, the project brings together skills you've learned in the program.

Q: What if I don't have the skills to solve the problem?

- > In some cases, you will need to learn new things
- > Your sponsor / mentor might be able to suggest resources

Ideally, the project brings together skills you've learned in the program.

Q: What if I don't have the skills to solve the problem?

- > In some cases, you will need to learn new things
- > Your sponsor / mentor might be able to suggest resources
- > Keep a list of open questions, skills needed
 - Break these down into manageable tasks (e.g., learn n-gram models)
- > You might distribute the work across teammates and share learnings

Sparse Feedback

Your mentor and sponsor should provide feedback on your work

Q: What if feedback isn't arriving often enough?

You can try:

- > Checking if there is a better medium for feedback (email, meetings, .
- > Asking if there is a better way to share results
 - Are questions clear enough?
 - Are things summarized?
(e.g., assemble separate questions into single slide)

Sparse Feedback

Your mentor and sponsor should provide feedback on your work

Q: What if feedback isn't arriving often enough?

You can try:

- > Understanding if there are busier/quieter periods
- > Working on aspects not requiring the feedback

Model Doesn't Generalize

You've built a model on training data
The performance on new data is worse

Q: Why does my model perform much worse on new data?

This is usually a symptom of overfitting. Can consider:

- k-fold cross validation (for ML models)
- Dropout layers (for deep neural networks)
- More suitable data (larger dataset, more recent data)
- Looking at the distribution of training data, new data.
Is there drift in distribution? If so, training set isn't sufficient.

Unstable Model

You've built a model

When you change hyperparameters slightly, the results change drastically

Q: Why is my model so unstable?

While we tune models for better performance,
a drastic change in output is not desirable

> The model may be too complex

Can try simpler model and examine sensitivity

> Did you cross validate the model? This is a good practice when feasible

Sponsor Unsure how to use Deliverables

Q: What if the sponsor is unsure how to use the deliverables?

- > Share work early and often. Avoid dropping everything on them at the end.
- > Leave time at the end to walk sponsor through deliverables
- > Have a meeting where sponsor team runs the code
- > Clearly document the work / code. A README file can be very effective. It can include objective, main functionality.

Final Thoughts

You have a support system to help you through capstones including:

- teammates
- sponsor
- faculty mentor
- Program Director (Adam Tashman)

Please lean on this system for help!